

Erik ERP System - Development Progress Report (10% Milestone)

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Executive Summary

We have successfully established the foundational architecture of Erik ERP, a comprehensive enterprise resource planning system built on React, TypeScript, and Lovable Cloud (Supabase backend). The current implementation represents approximately 10% of the complete system, focusing on core infrastructure, authentication framework, and essential module scaffolding.

1. Authentication & Authorization System

Implemented: - Basic Authentication Flow (email/password, session management, localStorage persistence, route protection, etc.) - User Profile Management (automatic creation, caching, and company association) - Role-Based Access Control (RBAC) with company-level data isolation and functions for super_admin and company roles - Multi-tenancy Architecture (trial setup, subscription tiers, and isolation) Basic Implementation: Current limitations include absence of OAuth, 2FA, password reset, and audit logs. These will be addressed in the next milestone (20–30%).

2. Database Architecture

Implemented Tables: Companies, Profiles, User Roles, Addon Modules, Employees, Departments, Attendance, Payroll, Customers, Invoices, Payments, Suppliers, Purchase Orders, Goods Receipts, and Inventory Items. Security: Row-Level Security enabled on all tables, company-scoped access, proper indexing, and automatic timestamp triggers.

3. Frontend Architecture

Core UI: Main layout with sidebar navigation, protected routes, dark/light mode, onboarding tour, and settings modules for company, users, roles, subscriptions, and guides. HR Module: Employee, Department, Attendance, Leave, Payroll, Loans, and Reports. Finance Module: Invoices, Payments, Chart of Accounts, and Journal Entry. Current Limitations: CRM, Procurement UI, and Employee Portal not yet implemented.

4. Implemented Features by Module

Human Resources: 60% complete – full CRUD, payroll, loans, attendance, and leave. Finance: 40% complete – chart of accounts, invoices, and payments. Inventory: 30% complete – item master and stock tracking. Procurement: 20% complete (backend only). Settings: 70% complete – user management, roles, add-ons, and subscriptions.

5. Technical Debt & Known Limitations

Missing features include OAuth, 2FA, session timeout, complex validation, caching, pagination, testing, and comprehensive documentation.

6. Next Steps (10% → 30% Roadmap)

Phase 2 Priorities: - Complete Procurement UI - Add Financial Reporting and CRM Module - Implement Enhanced Authentication - Build Employee Self-Service Portal - Develop Analytics Dashboards and Mobile Optimization

7. System Capabilities Summary

What Works: Multi-tenant setup, user authentication, HR operations, invoice management, inventory tracking, and company settings. Needs Work: Advanced analytics, integrations, workflow automation, mobile features, and performance optimization.

Conclusion

The Erik ERP system has established a solid architectural foundation. The current milestone represents infrastructure completion, setting the stage for rapid module development and scale-ready expansion.